Food-Delivery-SQL-Analysis-using-AZURE-SQL

Objective:

The main objective of this project is to analyze food delivery data using SQL. This project focuses on understanding customer orders, restaurant performance, popular food items, and revenue. SQL queries are used to extract meaningful business insights and support better decision-making.

Database

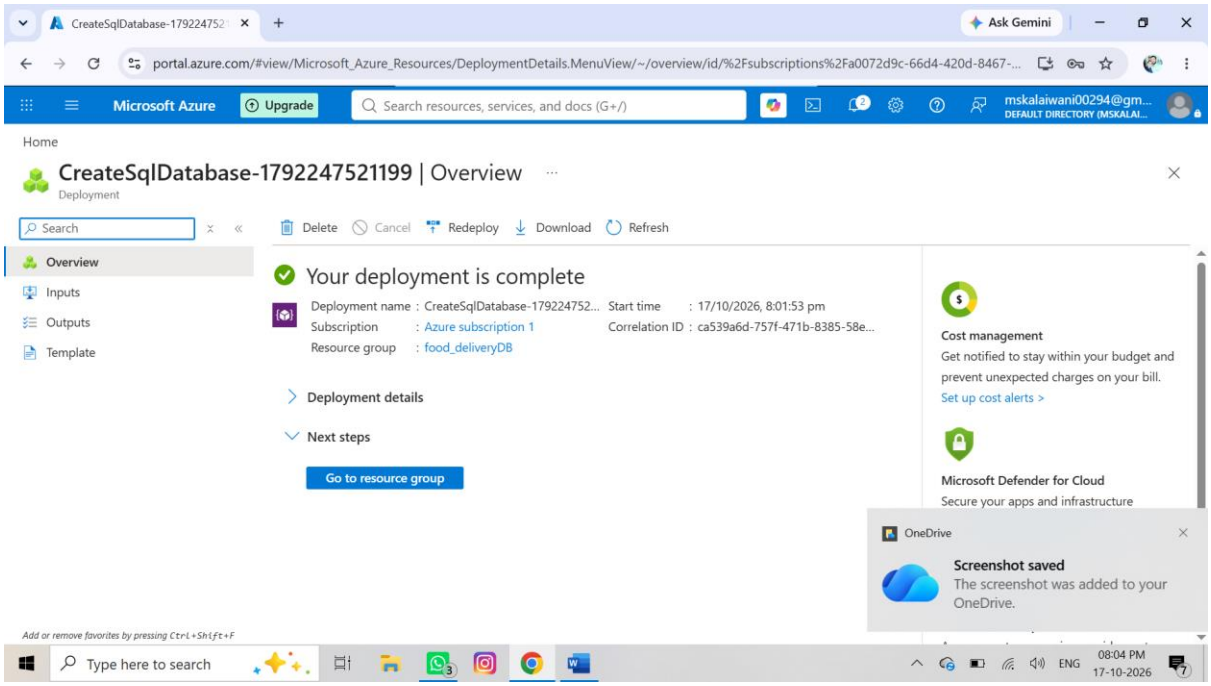
The project uses an Azure SQL Database named "food-delivery-DB".

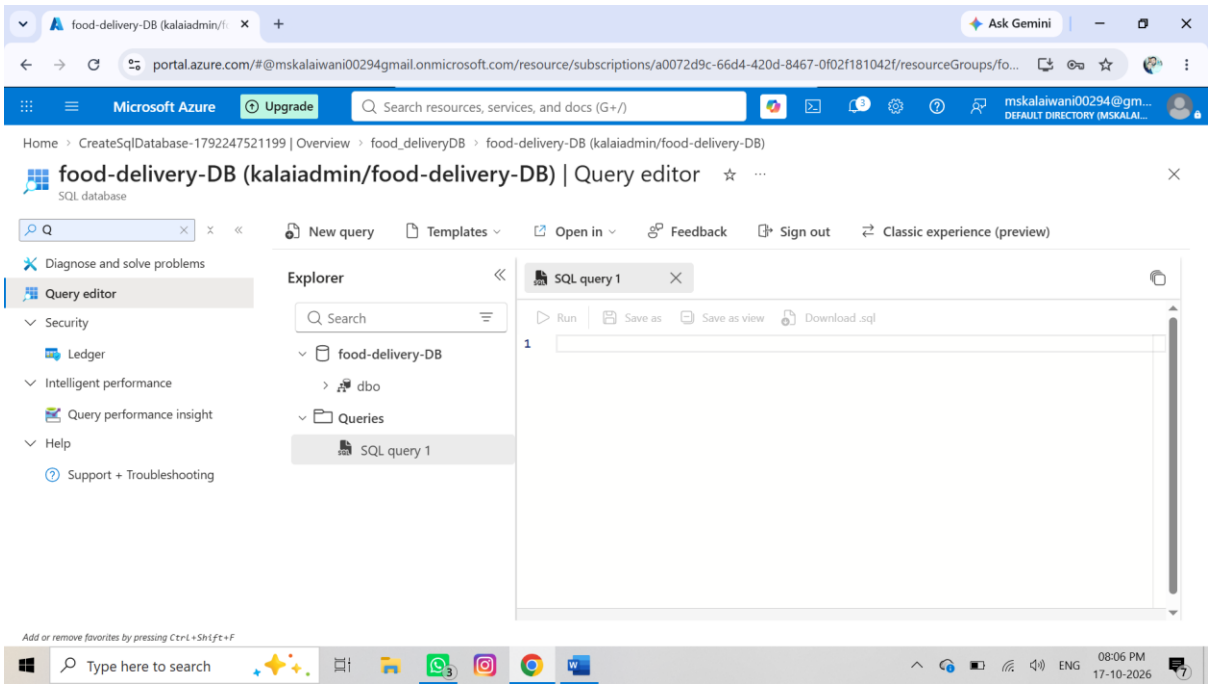
The database contains five tables:

1. Customers: Stores customer details such as name, phone number, email, and city.
2. Restaurants: Stores restaurant names, locations, and ratings.
3. Food_Items: Contains food names, prices, and restaurant details.
4. Orders: Stores order dates, customer details, order status, and total amounts.
5. Order_Items: Contains food items, quantities, and item prices for each order.

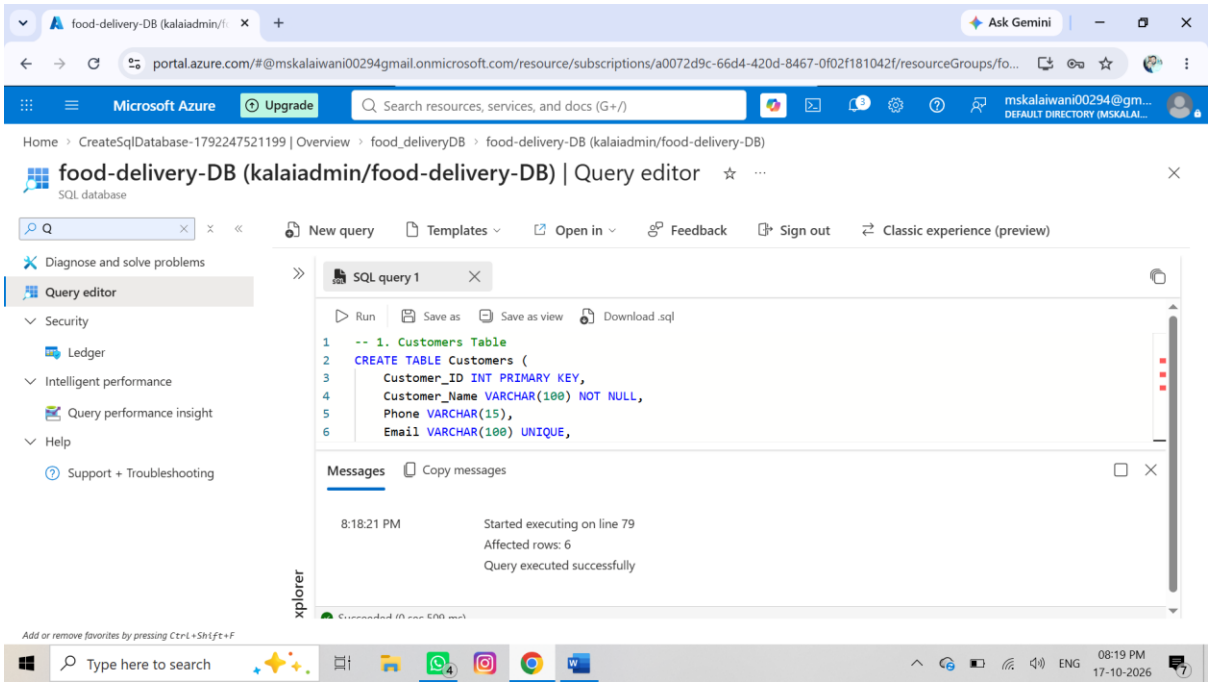
SQL Concepts Used:

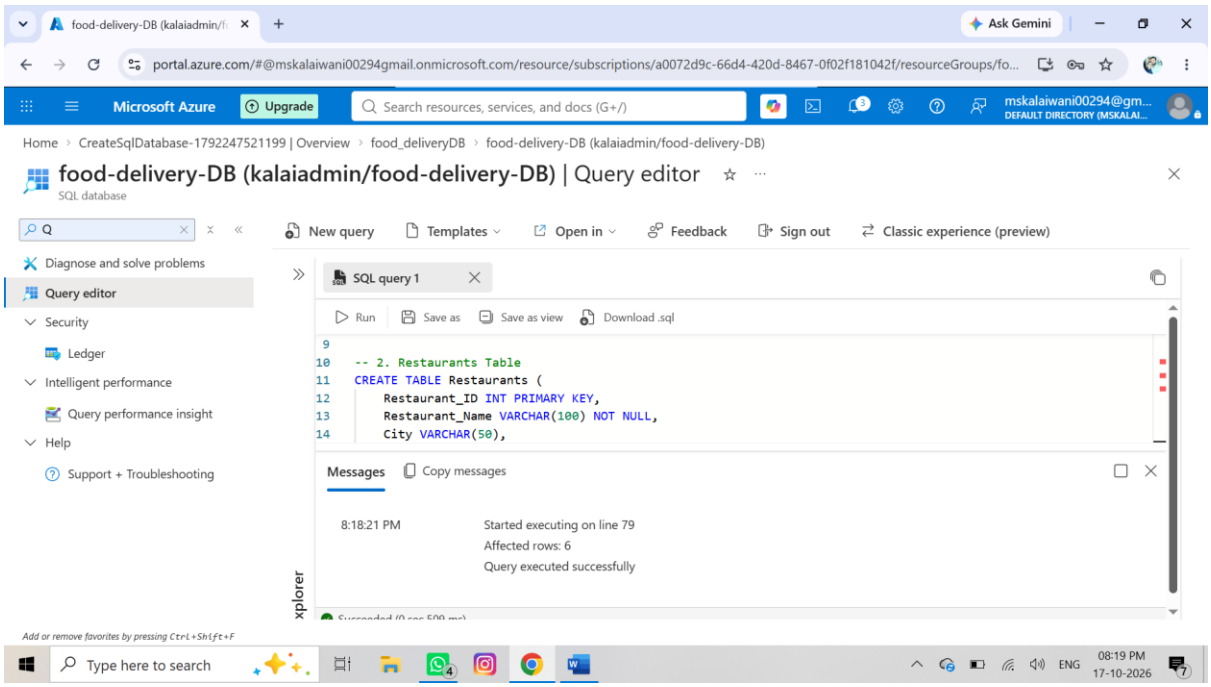
a. CREATE TABLE
b. PRIMARY KEY and FOREIGN KEY
c. INSERT and SELECT
d. Aggregate functions: COUNT(), SUM()
e. JOIN operations
f. GROUP BY and ORDER BY
g. TOP clause

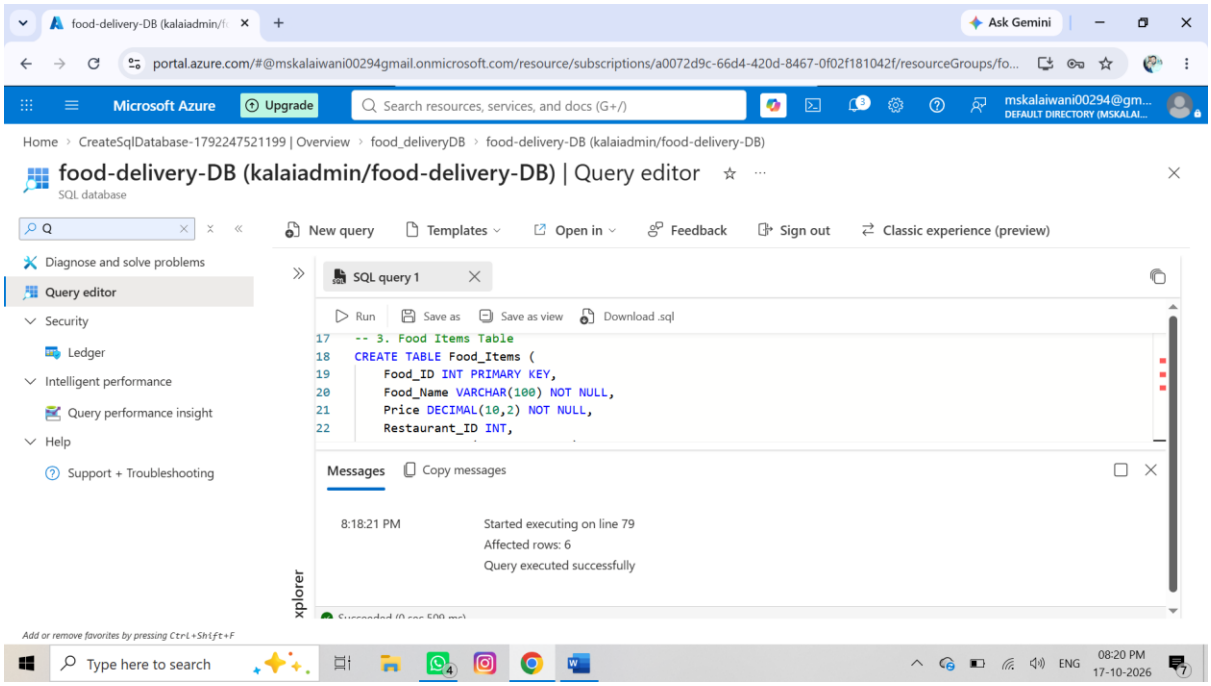


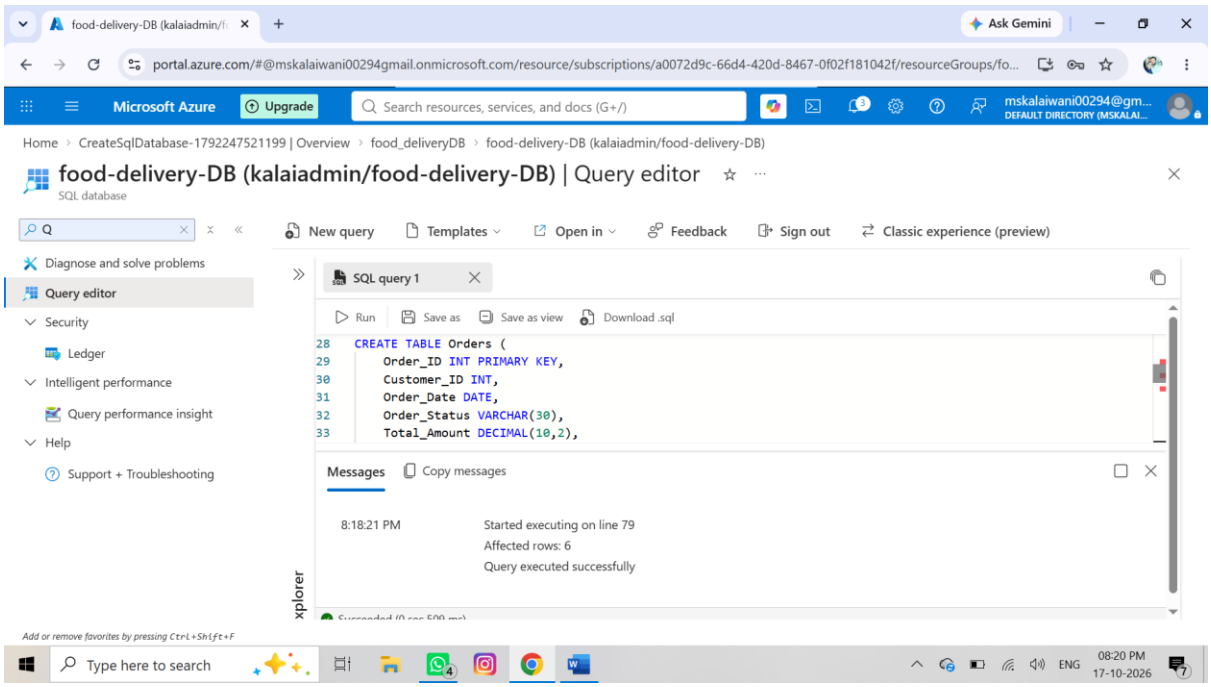


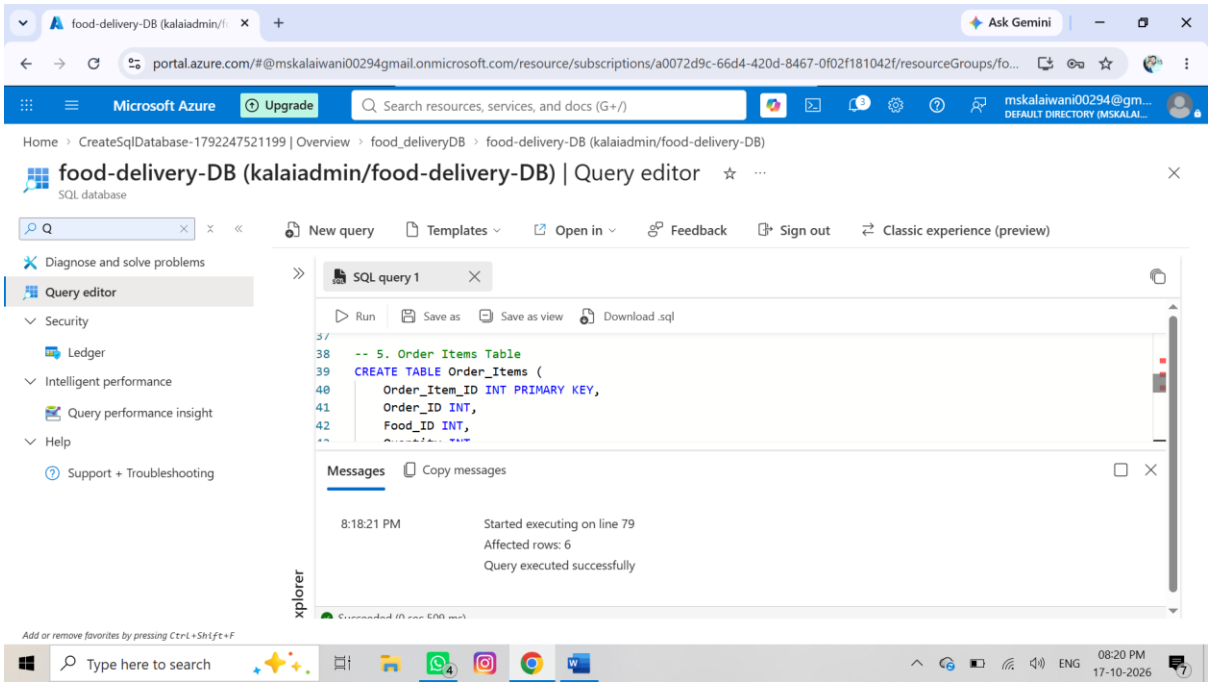
Create database query:

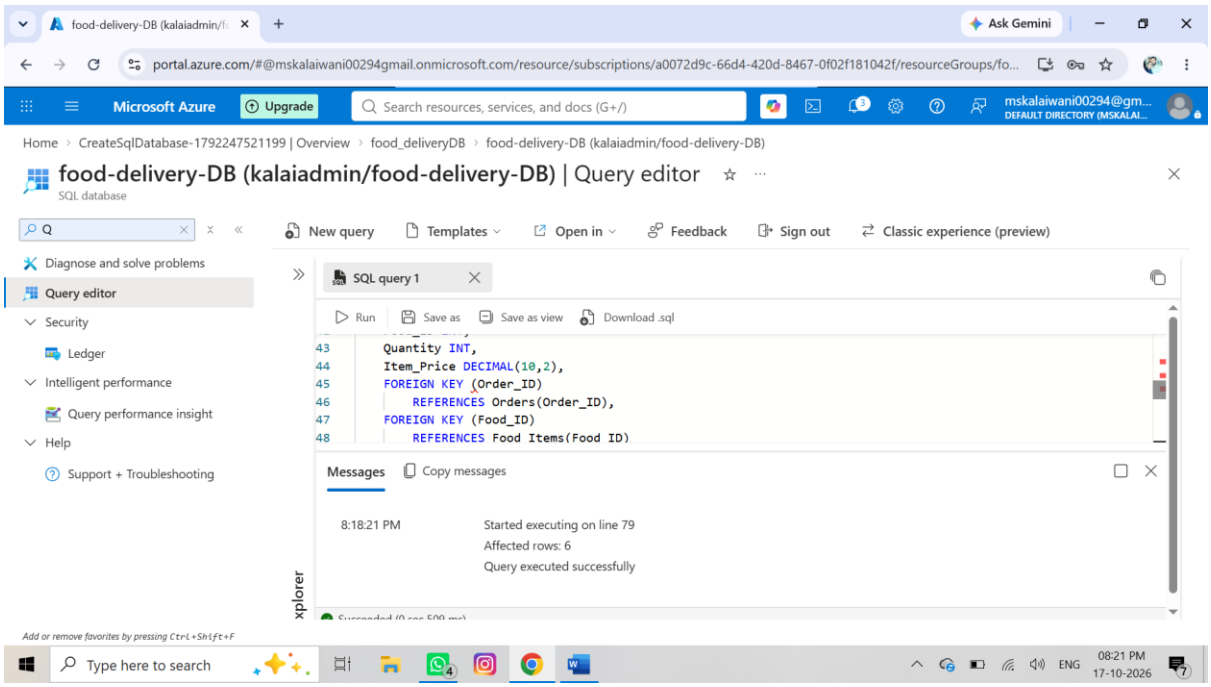

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SQL query 1
Run
Save as
Save as view
Download .sql
1 -- 1. Customers Table
2 CREATE TABLE Customers (
3 Customer_ID INT PRIMARY KEY,
4 Customer_Name VARCHAR(100) NOT NULL,
5 Phone VARCHAR(15),
6 Email VARCHAR(100) UNIQUE,
Messages
Copy messages
8:18:21 PM
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Affected rows: 6
Query executed successfully
Add or remove favorites by pressing Ctrl+Shift+F
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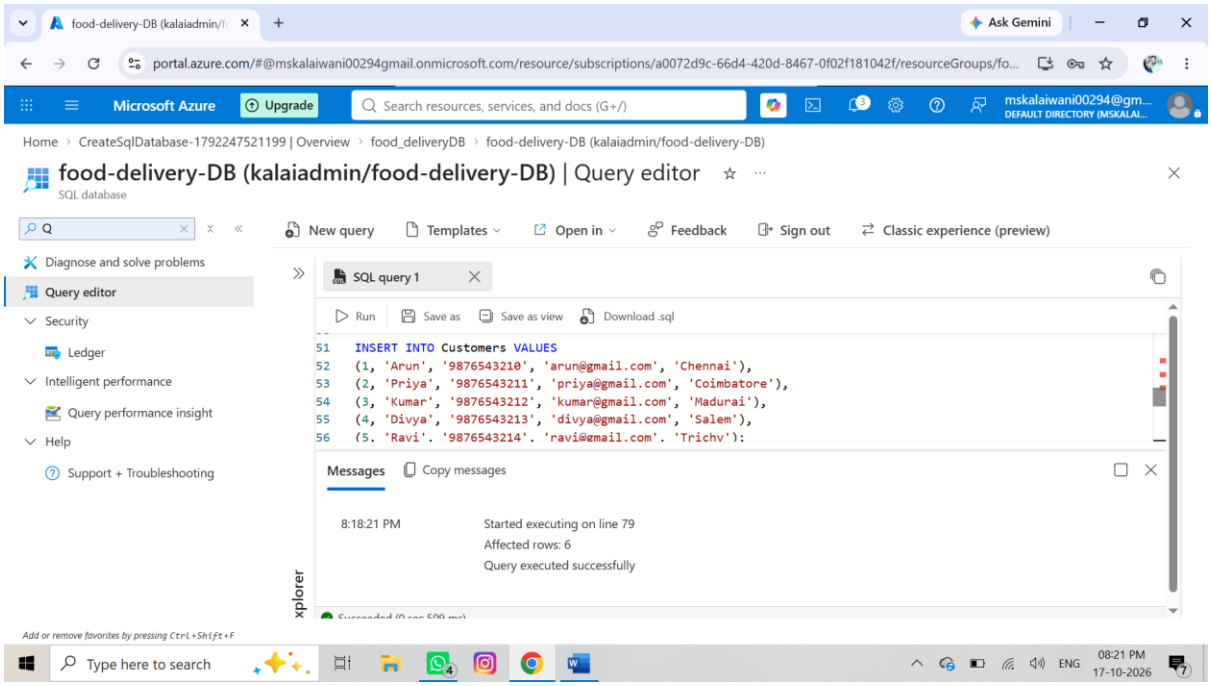

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SQL query 1
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Save as view
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9
10 -- 2. Restaurants Table
11 CREATE TABLE Restaurants (
12 Restaurant_ID INT PRIMARY KEY,
13 Restaurant_Name VARCHAR(100) NOT NULL,
14 City VARCHAR(50),
Messages
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Started executing on line 79
Affected rows: 6
Query executed successfully
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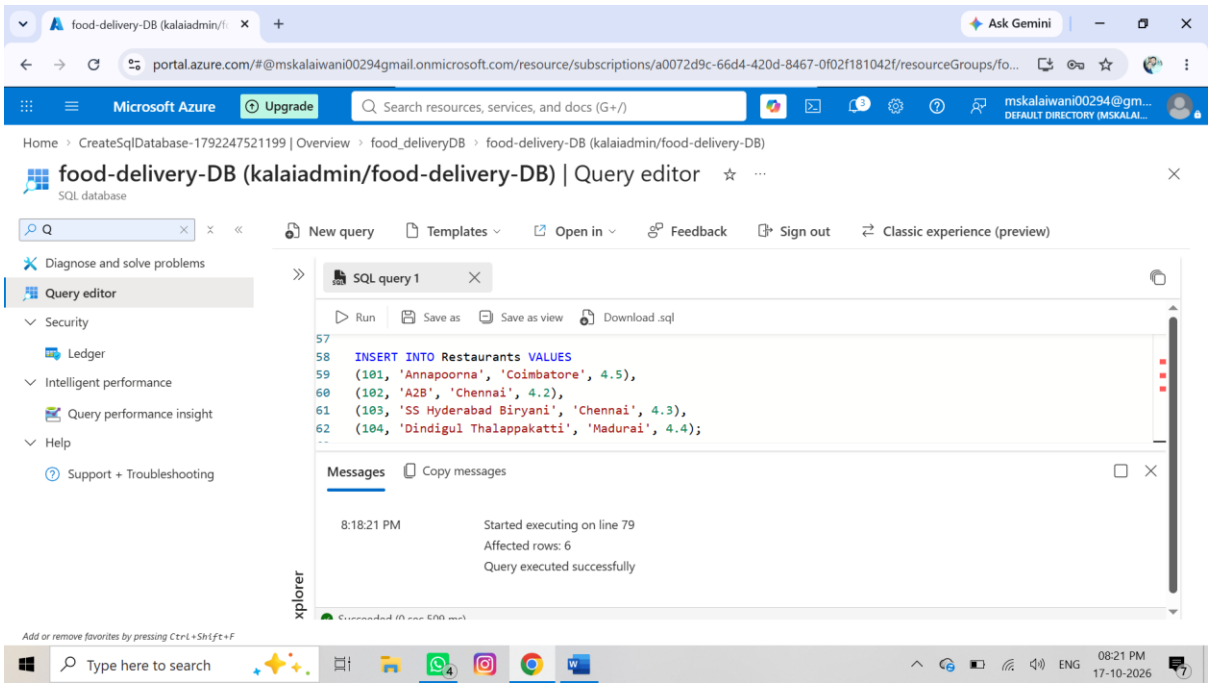

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Run
Save as
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17 -- 3. Food Items Table
18 CREATE TABLE Food_Items (
19 Food_ID INT PRIMARY KEY,
20 Food_Name VARCHAR(100) NOT NULL,
21 Price DECIMAL(10,2) NOT NULL,
22 Restaurant_ID INT,
Messages
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Started executing on line 79
Affected rows: 6
Query executed successfully
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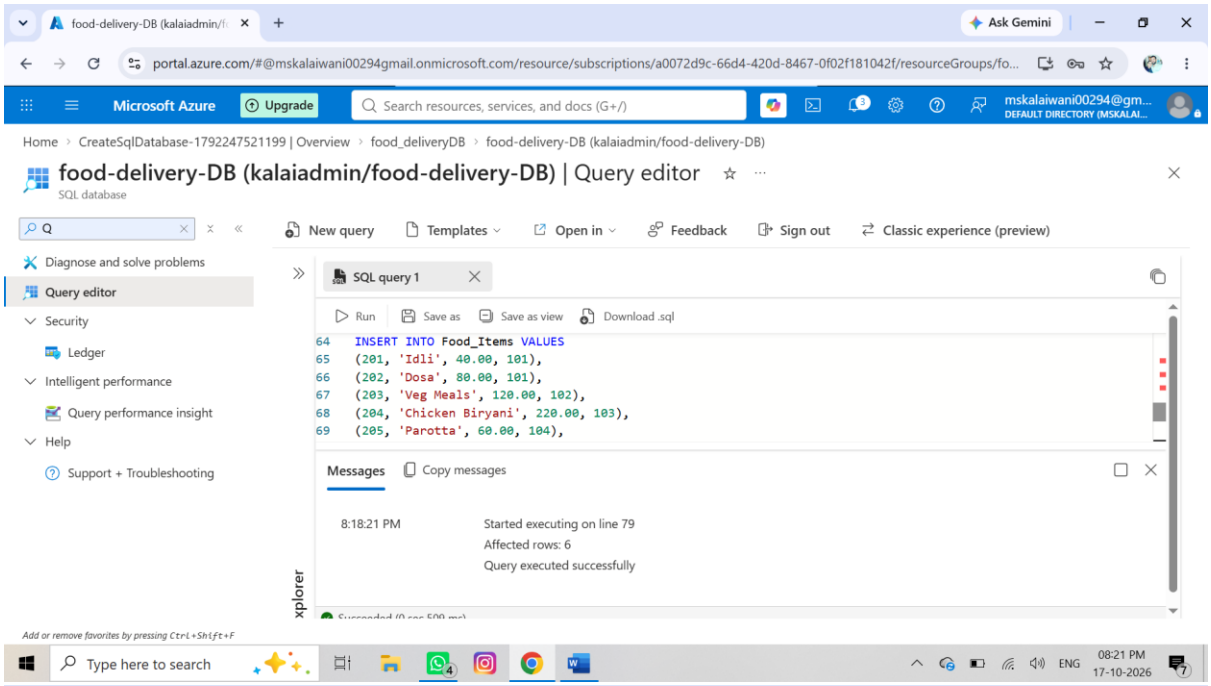

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SQL query 1
Run
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28 CREATE TABLE Orders (
29 Order_ID INT PRIMARY KEY,
30 Customer_ID INT,
31 Order_Date DATE,
32 Order_Status VARCHAR(30),
33 Total_Amount DECIMAL(10,2),
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
Query executed successfully
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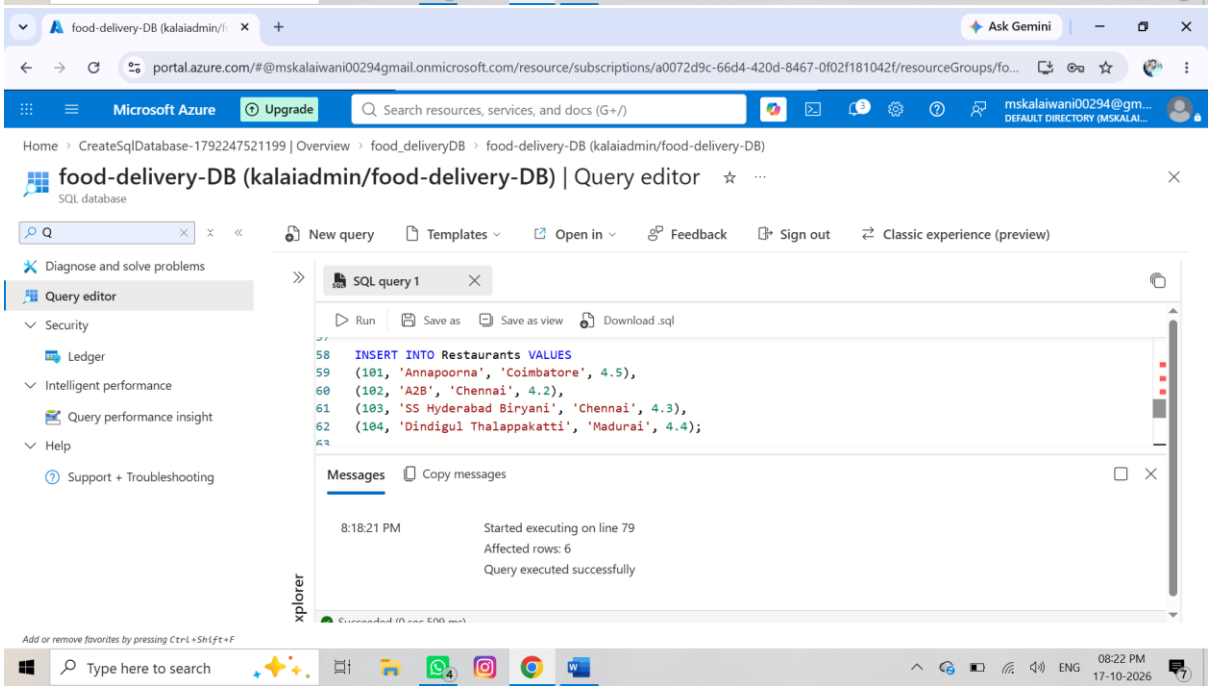

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Run
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38 -- 5. Order Items Table
39 CREATE TABLE Order_Items (
40 Order_Item_ID INT PRIMARY KEY,
41 Order_ID INT,
42 Food_ID INT,
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
Query executed successfully
Add or remove favorites by pressing Ctrl+Shift+F
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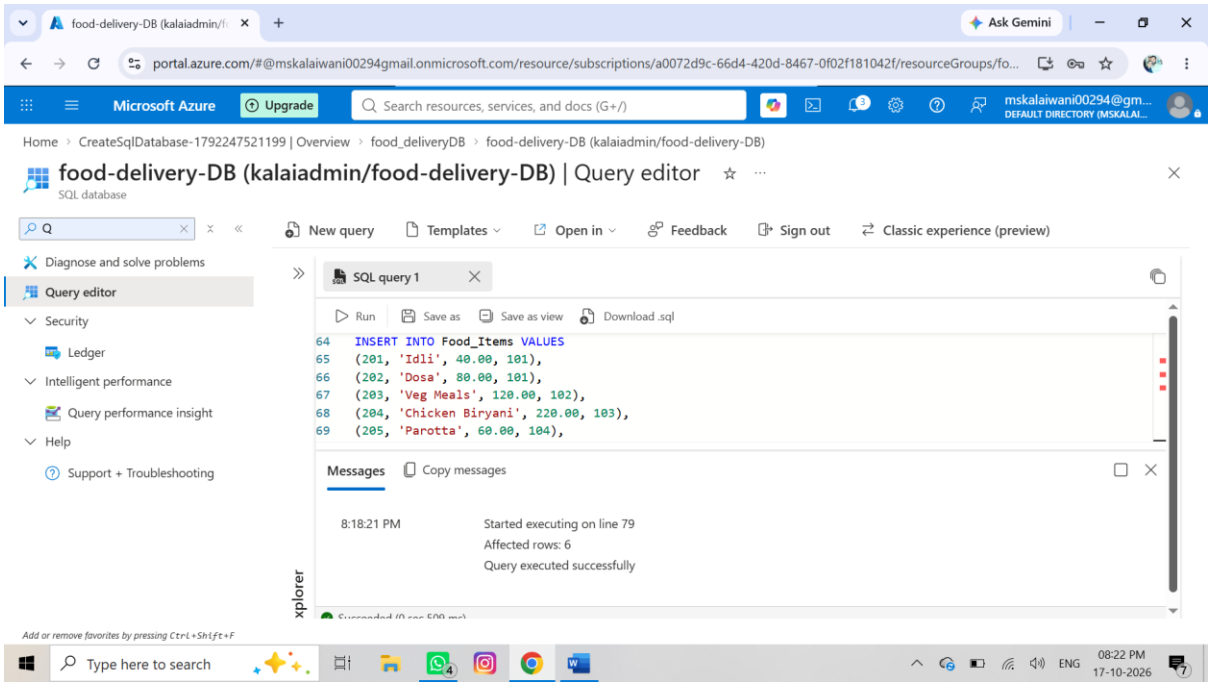

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43 Quantity INT,
44 Item_Price DECIMAL(10,2),
45 FOREIGN KEY (Order_ID)
46 REFERENCES Orders(Order_ID),
47 FOREIGN KEY (Food_ID)
48 REFERENCES Food Items(Food ID)
Messages
Copy messages
8:18:21 PM
Started executing on line 79
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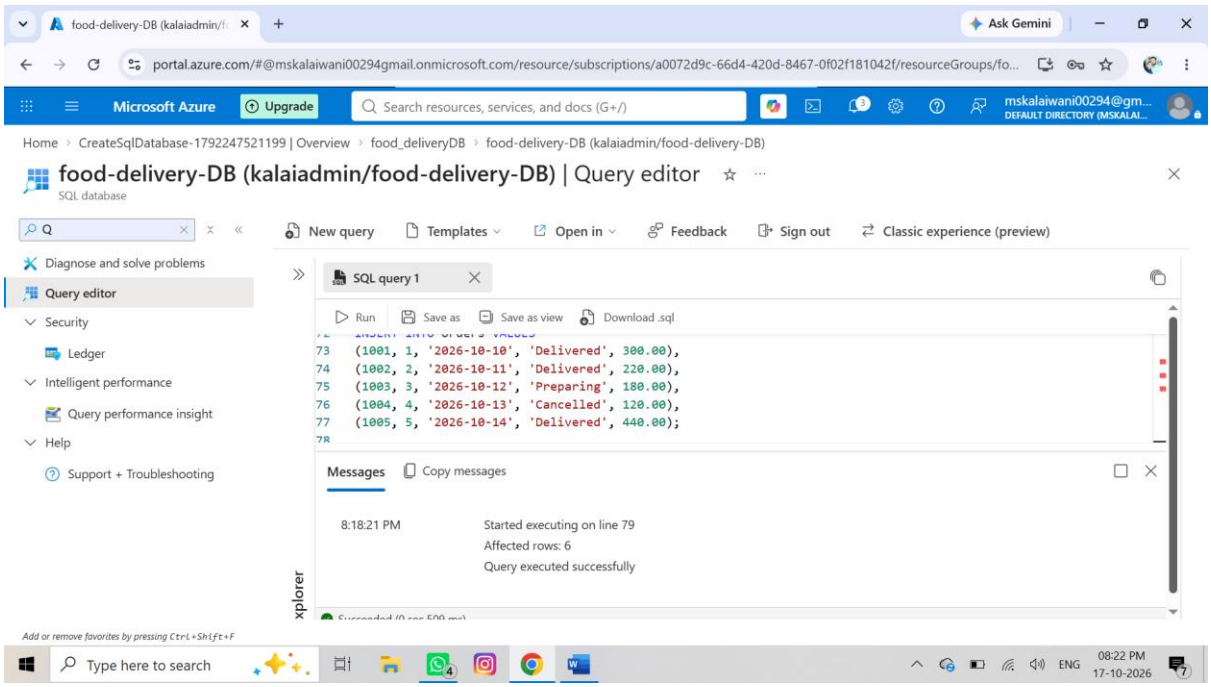

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51 INSERT INTO Customers VALUES
52 (1, 'Arun', '9876543210', 'arun@gmail.com', 'Chennai'),
53 (2, 'Priya', '9876543211', 'priya@gmail.com', 'Coimbatore'),
54 (3, 'Kumar', '9876543212', 'kumar@gmail.com', 'Madurai'),
55 (4, 'Divya', '9876543213', 'divya@gmail.com', 'Salem'),
56 (5, 'Ravi', '9876543214', 'ravi@gmail.com', 'Trichy');
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
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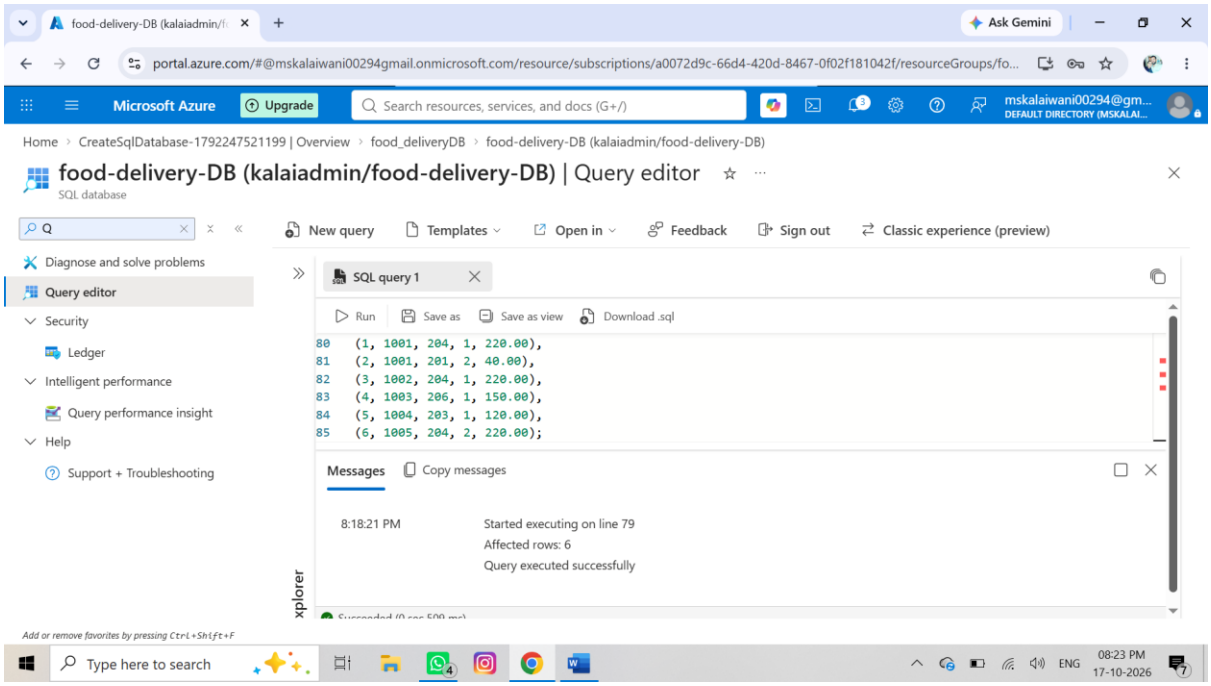

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SQL query 1
Run
Save as
Save as view
Download .sql
57
58 INSERT INTO Restaurants VALUES
59 (101, 'Annapoorna', 'Coimbatore', 4.5),
60 (102, 'A2B', 'Chennai', 4.2),
61 (103, 'SS Hyderabad Biryani', 'Chennai', 4.3),
62 (104, 'Dindigul Thalappakatti', 'Madurai', 4.4);
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
Query executed successfully
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SQL query 1
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64 INSERT INTO Food_Items VALUES
65 (201, 'Idli', 40.00, 101),
66 (202, 'Dosa', 80.00, 101),
67 (203, 'Veg Meals', 120.00, 102),
68 (204, 'Chicken Biryani', 220.00, 103),
69 (205, 'Parotta', 60.00, 104),
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
Query executed successfully
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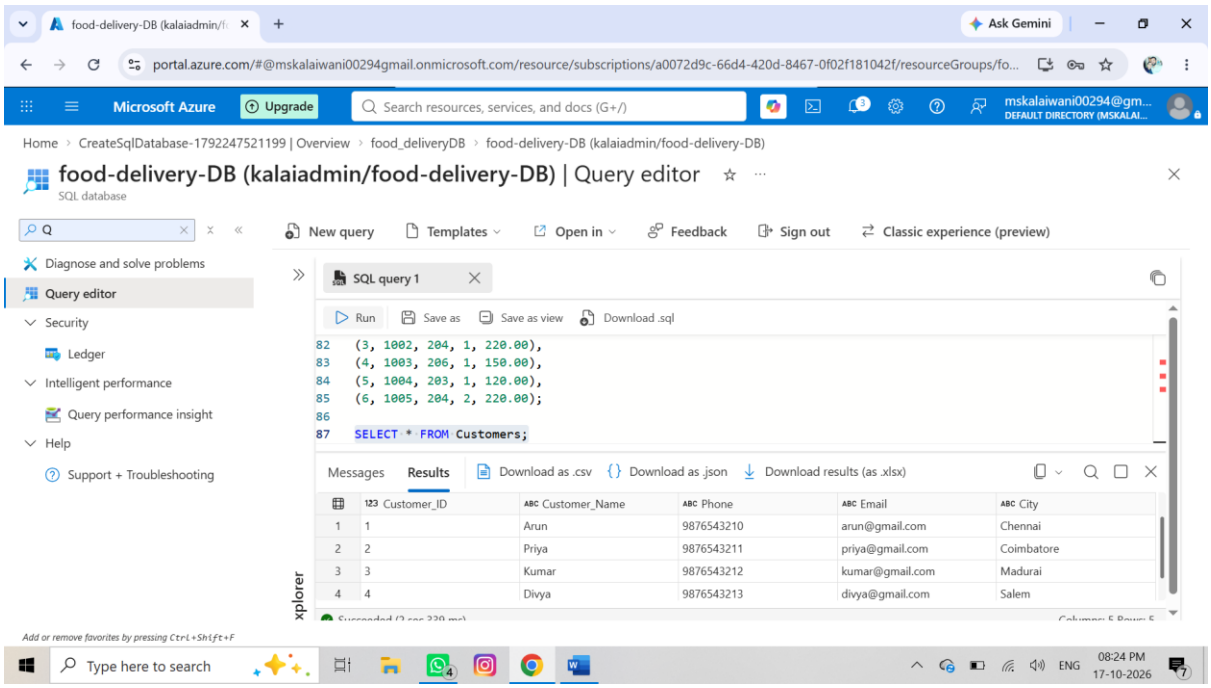

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SQL query 1
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58 INSERT INTO Restaurants VALUES
59 (101, 'Annapoorna', 'Coimbatore', 4.5),
60 (102, 'A2B', 'Chennai', 4.2),
61 (103, 'SS Hyderabad Biryani', 'Chennai', 4.3),
62 (104, 'Dindigul Thalappakatti', 'Madurai', 4.4);
Messages
Copy messages
8:18:21 PM
Started executing on line 79
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Query executed successfully
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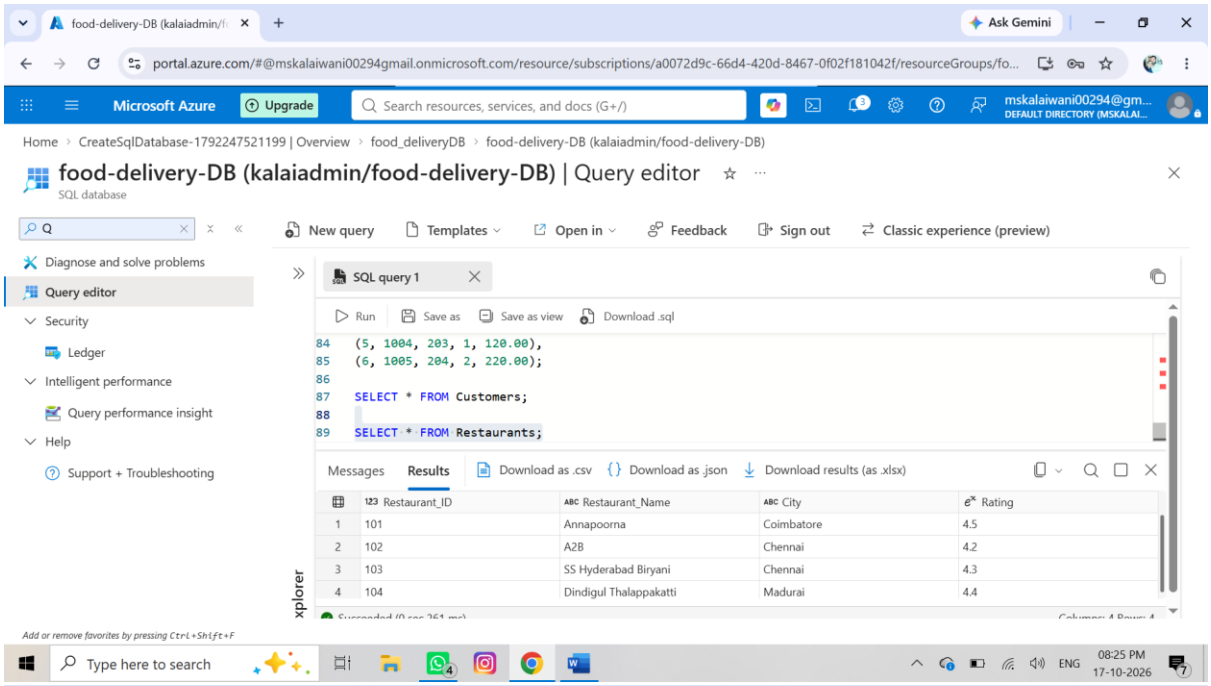

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65 (201, 'Idli', 40.00, 101),
66 (202, 'Dosa', 80.00, 101),
67 (203, 'Veg Meals', 120.00, 102),
68 (204, 'Chicken Biryani', 220.00, 103),
69 (205, 'Parotta', 60.00, 104),
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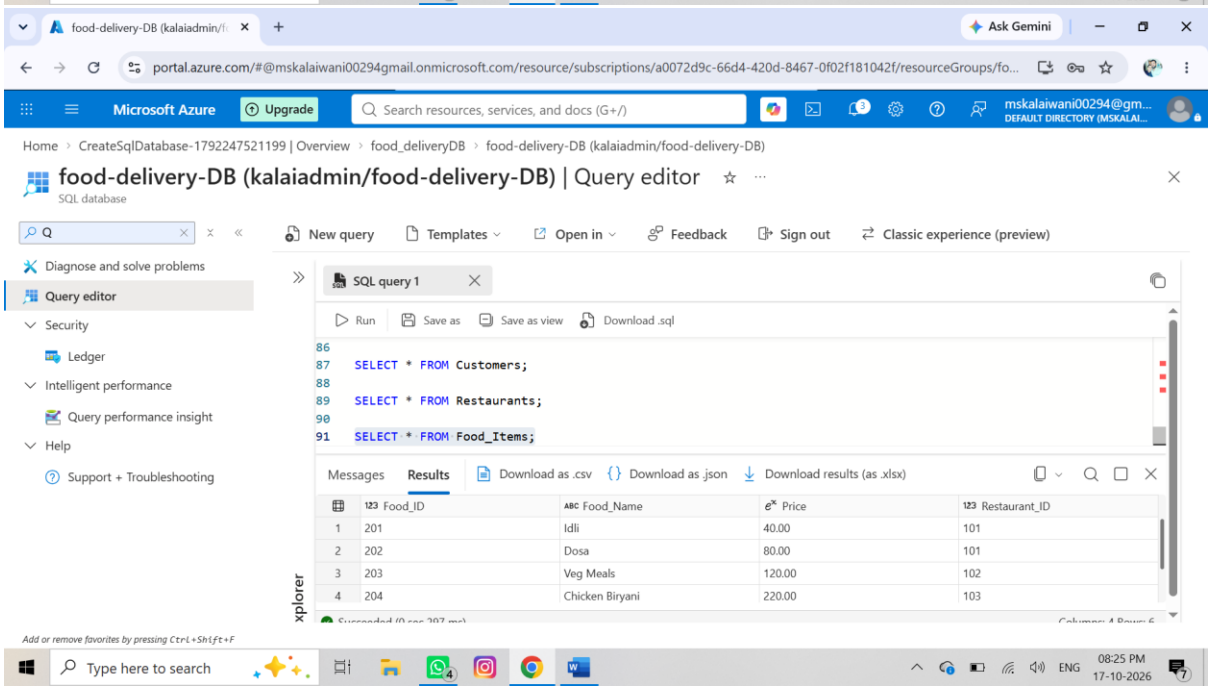

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SQL query 1
Run
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73 (1001, 1, '2026-10-10', 'Delivered', 300.00),
74 (1002, 2, '2026-10-11', 'Delivered', 220.00),
75 (1003, 3, '2026-10-12', 'Preparing', 180.00),
76 (1004, 4, '2026-10-13', 'Cancelled', 120.00),
77 (1005, 5, '2026-10-14', 'Delivered', 440.00);
78
Messages
Copy messages
8:18:21 PM
Started executing on line 79
Affected rows: 6
Query executed successfully
xplorer
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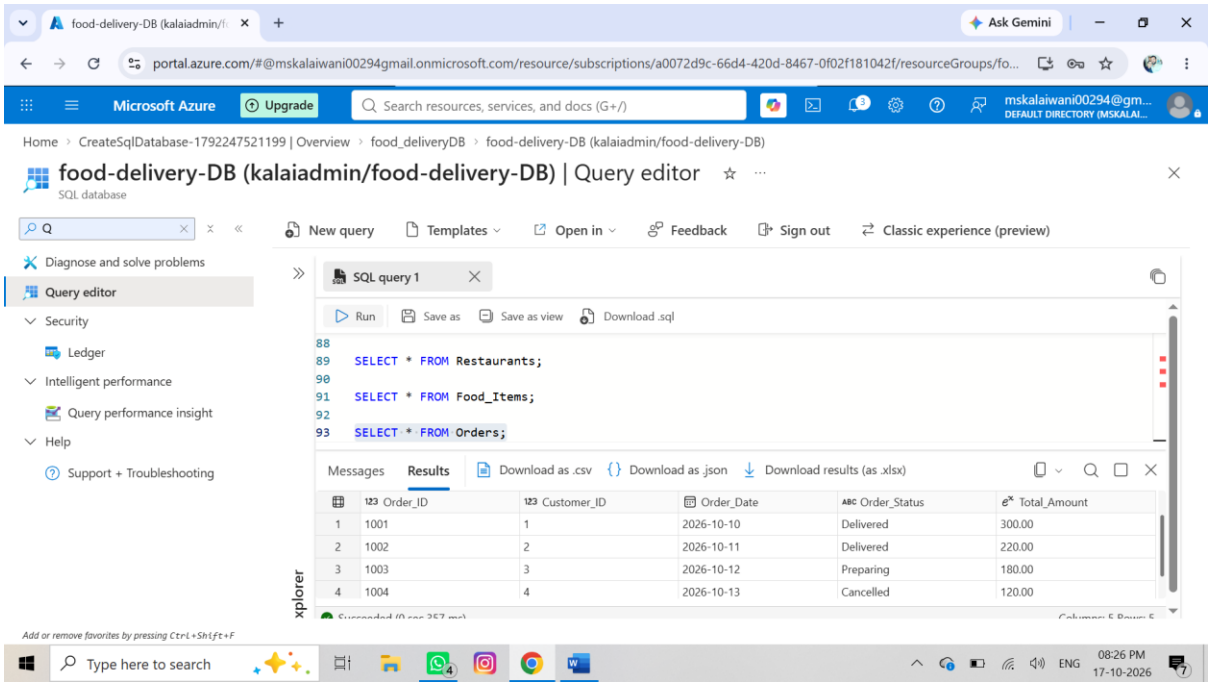


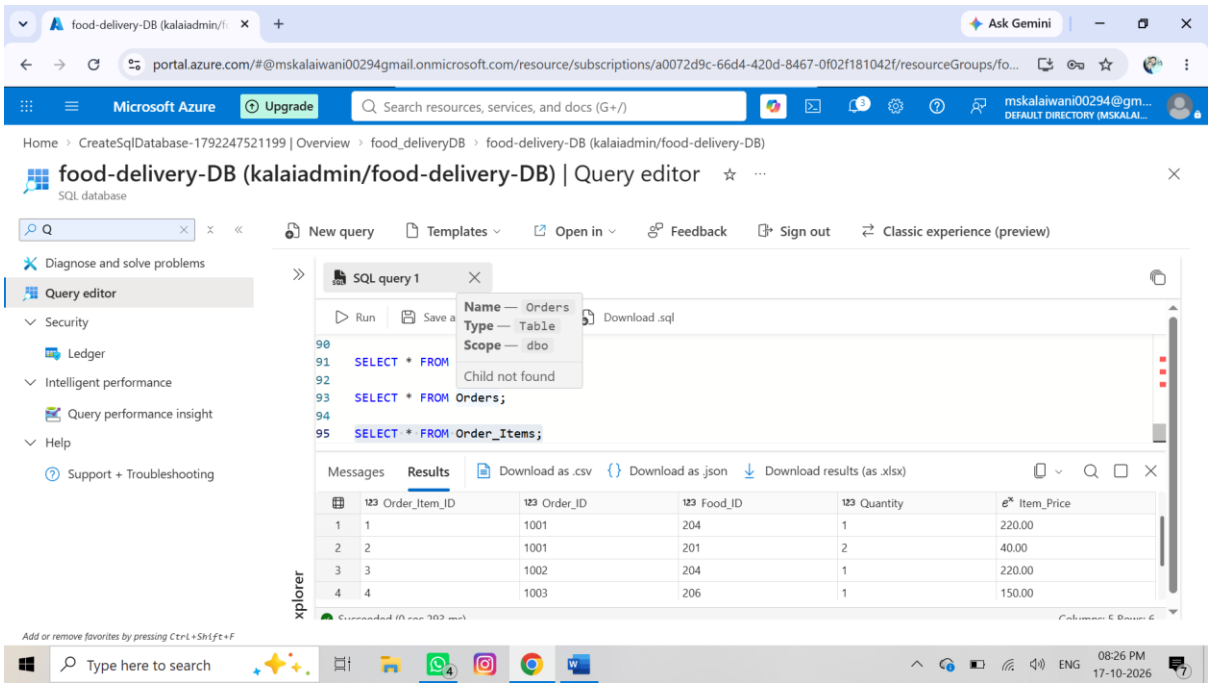
Display All data:




food-delivery-DB (kalaiadmin/f
Ask Gemini
portal.azure.com/#@mskalaiwani00294gmail.onmicrosoft.com/resource/subscriptions/a0072d9c-66d4-420d-8467-0f02f181042f/resourceGroups/fo...
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DEFAULT DIRECTORY (MSKALAI...
Home > CreateSqlDatabase-1792247521199 | Overview > food_deliveryDB > food-delivery-DB (kalaiadmin/food-delivery-DB)
food-delivery-DB (kalaiadmin/food-delivery-DB) | Query editor
SQL database
New query
Templates
Open in
Feedback
Sign out
Classic experience (preview)
Diagnose and solve problems
Query editor
Security
Ledger
Intelligent performance
Query performance insight
Help
Support + Troubleshooting
SQL query 1
Run
Save as
Save as view
Download .sql
84 (5, 1004, 203, 1, 120.00),
85 (6, 1005, 204, 2, 220.00);
86
87 SELECT * FROM Customers;
88
89 SELECT * FROM Restaurants;
Messages
Results
Download as .csv
Download as .json
Download results (as .xlsx)
Restaurant_ID
Restaurant_Name
City
Rating
1 101 Annapoorna Coimbatore 4.5
2 102 A2B Chennai 4.2
3 103 SS Hyderabad Biryani Chennai 4.3
4 104 Dindigul Thalappakatti Madurai 4.4
Add or remove favorites by pressing Ctrl+Shift+F
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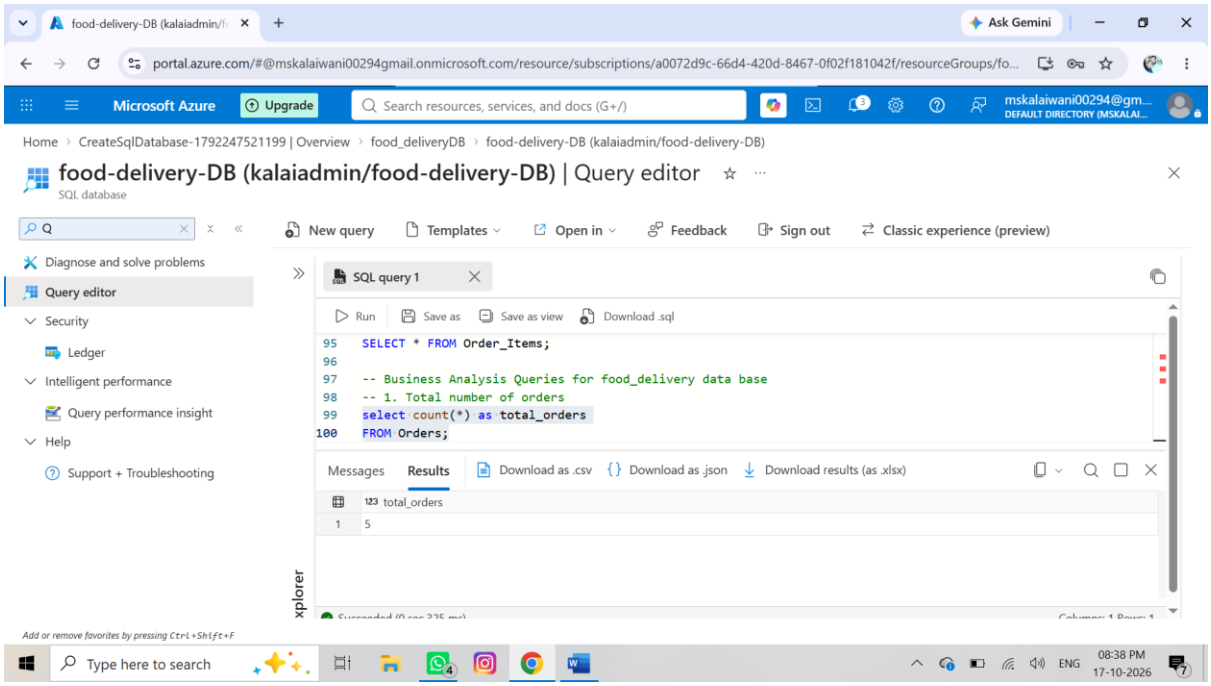

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Intelligent performance
Query performance insight
Help
Support + Troubleshooting
SQL query 1
Run
Save as
Save as view
Download .sql
86
87 SELECT * FROM Customers;
88
89 SELECT * FROM Restaurants;
90
91 SELECT * FROM Food_Items;
Messages
Results
Download as .csv
Download as .json
Download results (as .xlsx)
Food_ID
Food_Name
Price
Restaurant_ID
1 201 Idli 40.00 101
2 202 Dosa 80.00 101
3 203 Veg Meals 120.00 102
4 204 Chicken Biryani 220.00 103
Add or remove favorites by pressing Ctrl+Shift+F
Type here to search
ENG
08:25 PM
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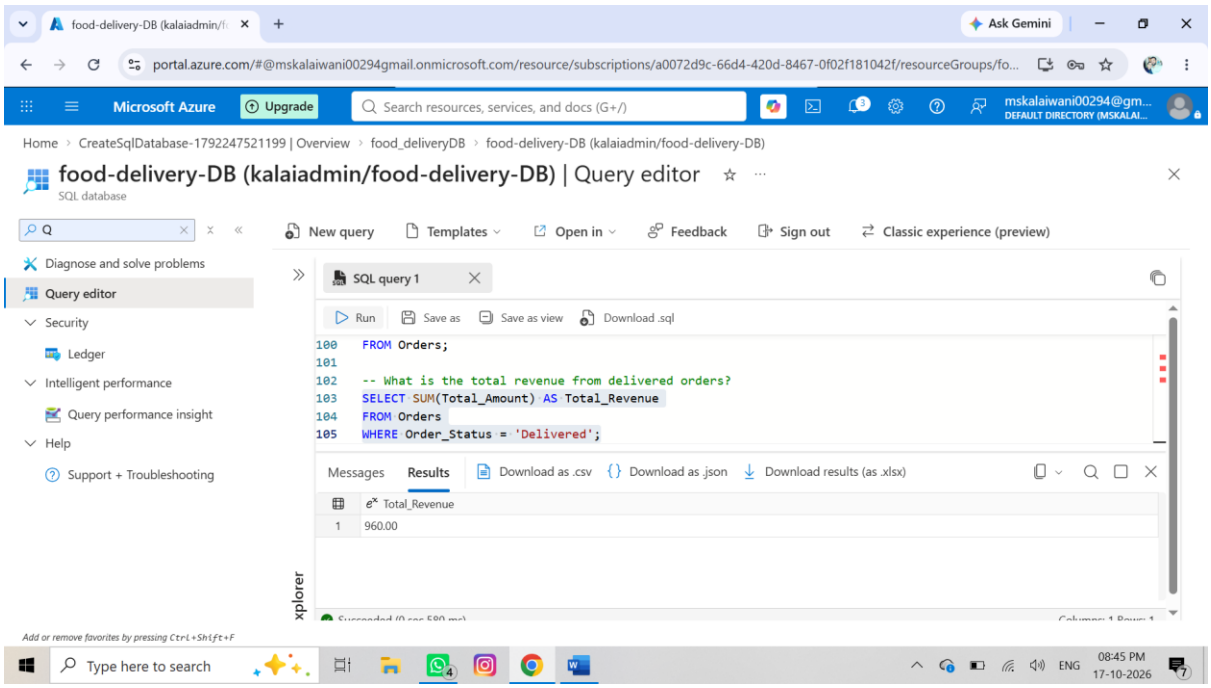


Business Analysis Queries for food_delivery data base

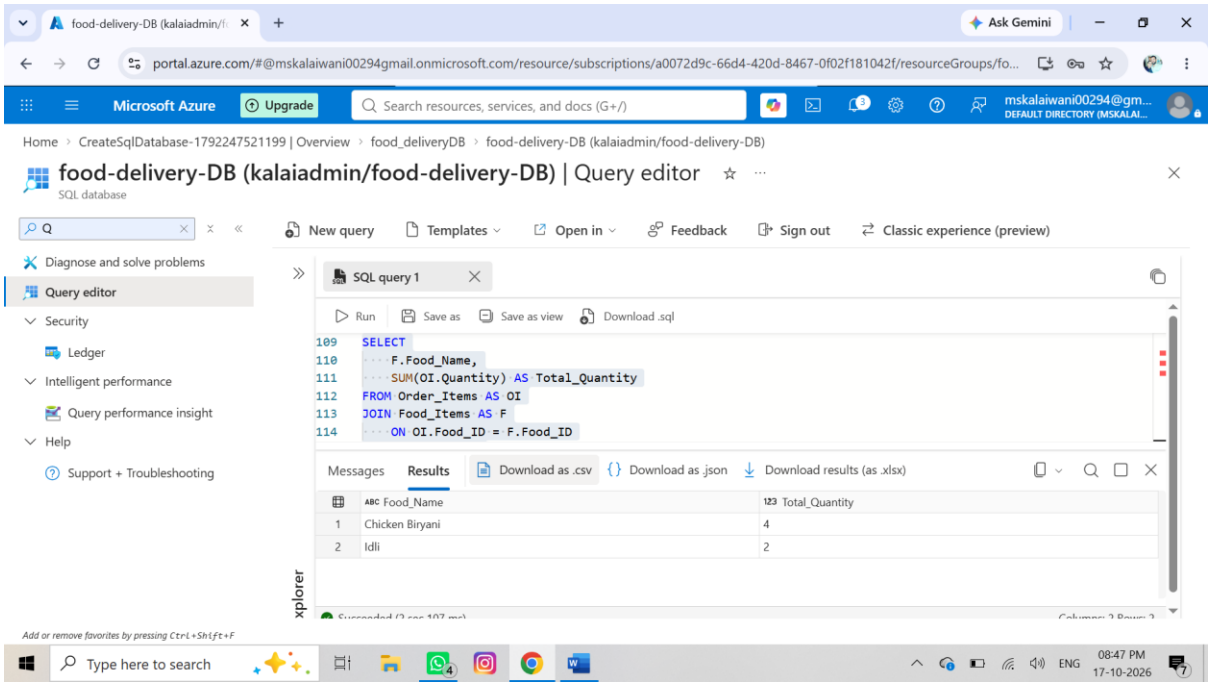
1. Total number of orders



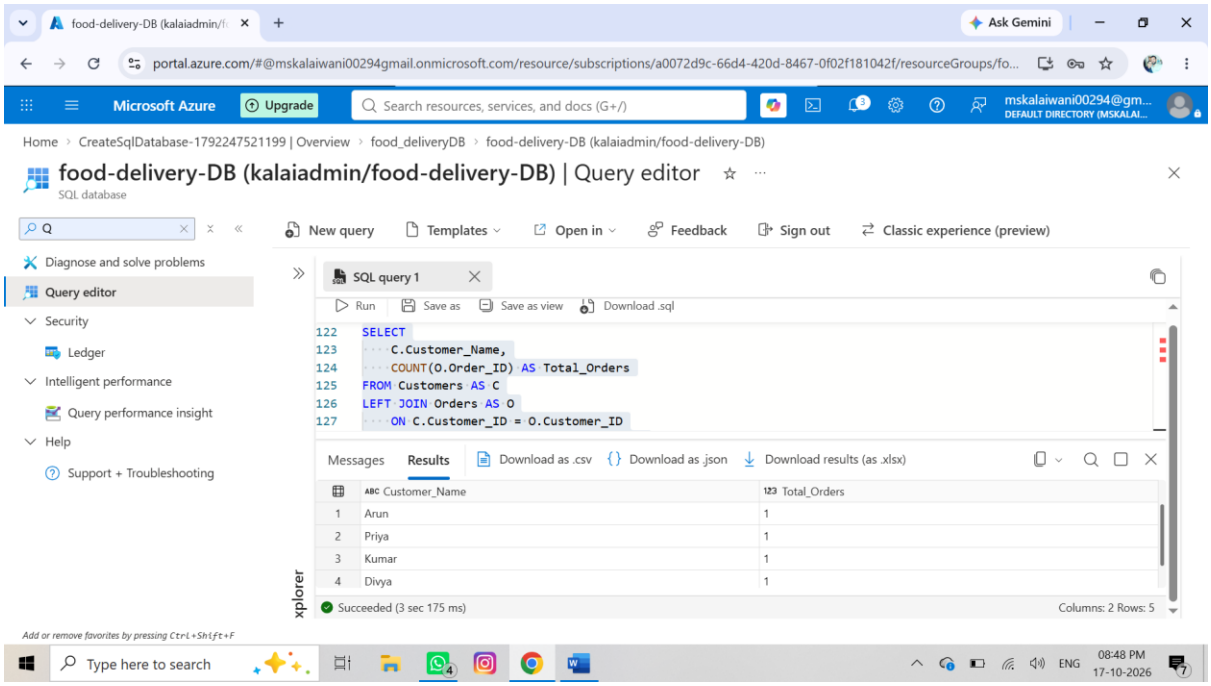
What is the total revenue from delivered orders?



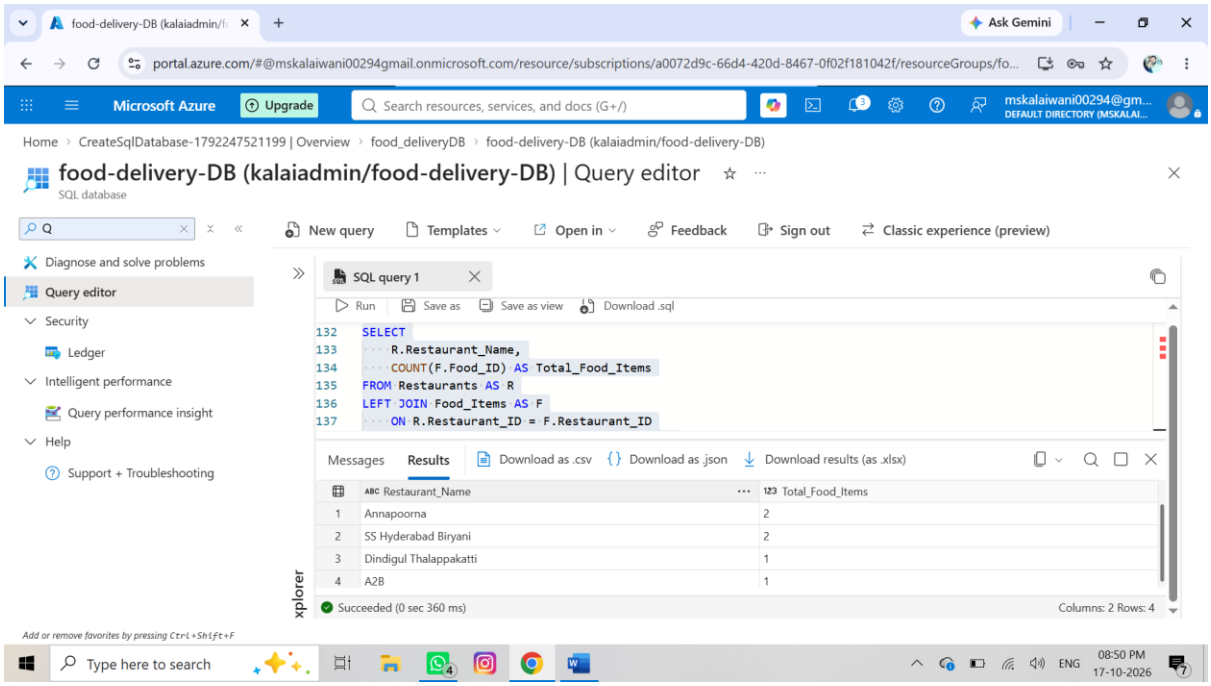
3. Which food items are the most popular?



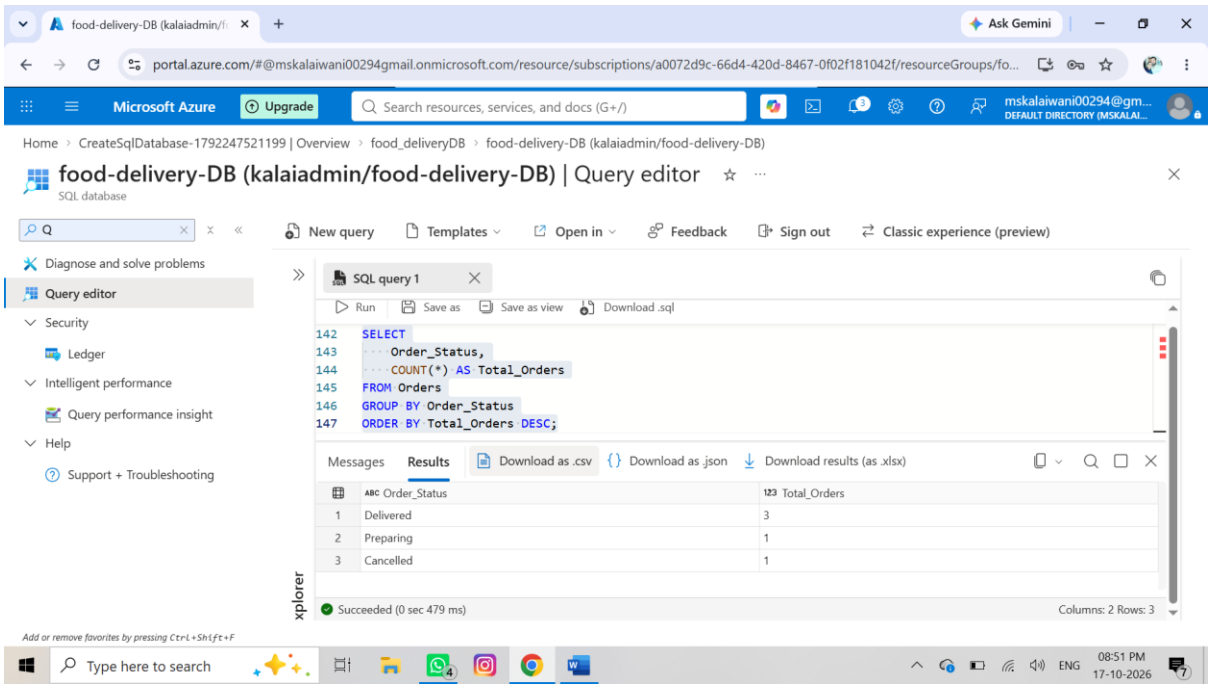
Which customers place the most orders?



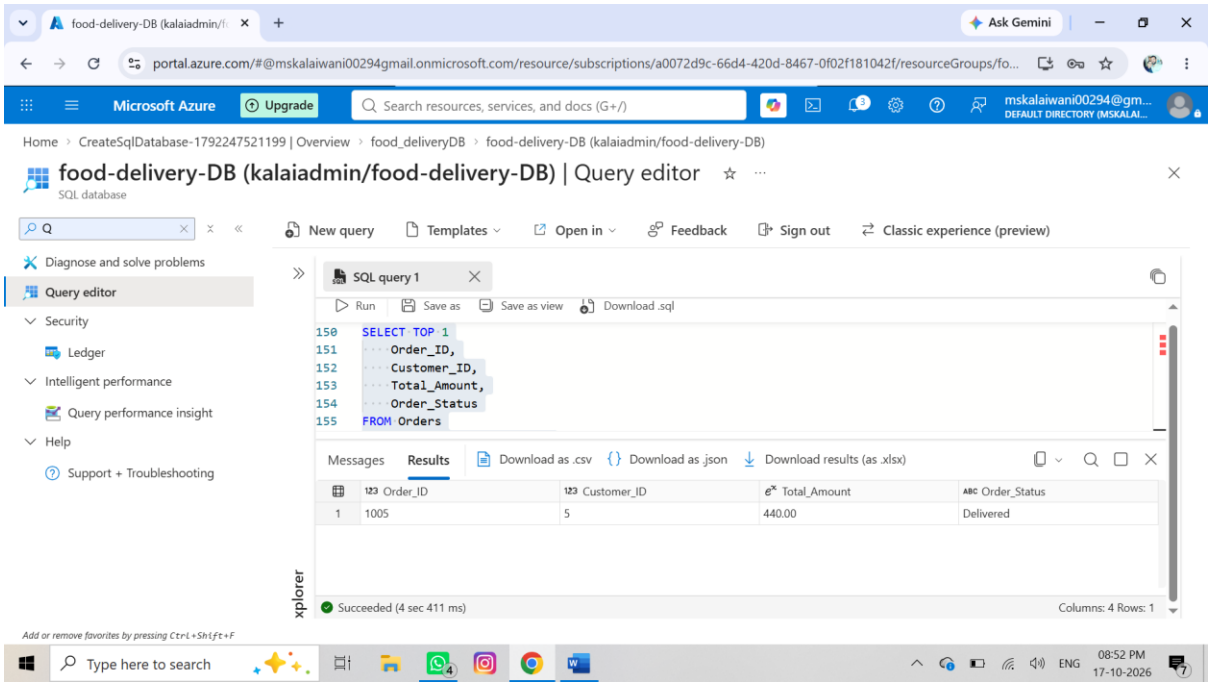
Which restaurants offer the most food items?



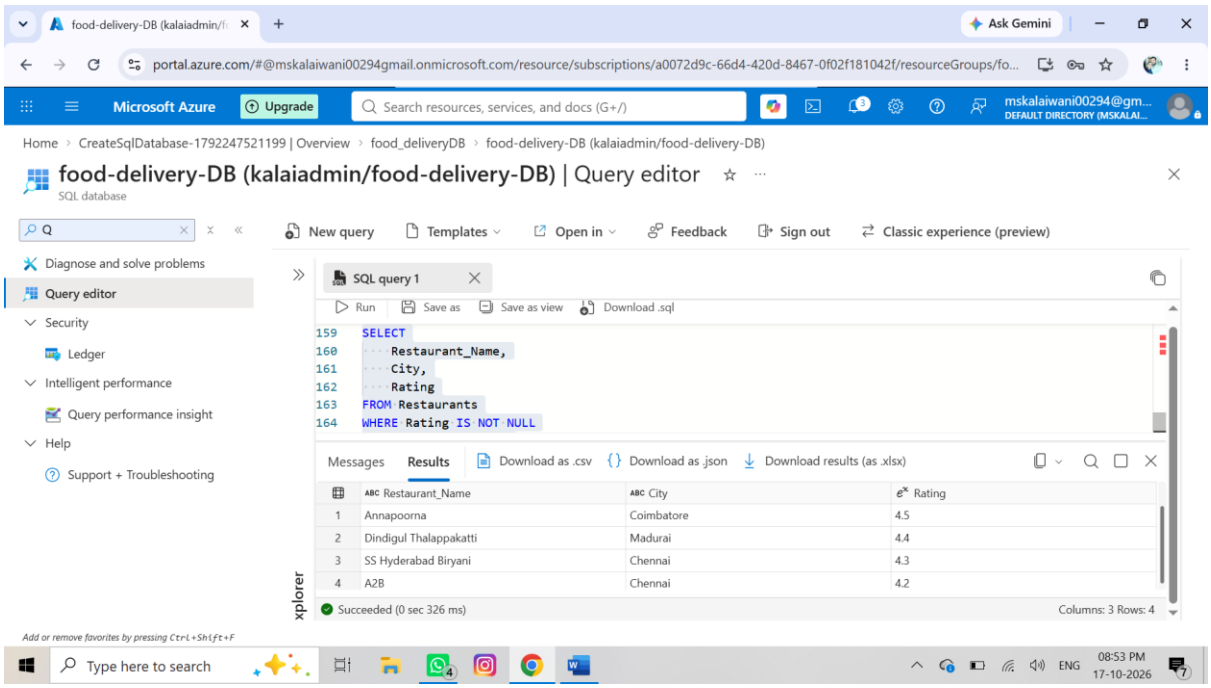
What is the order count for each order status?

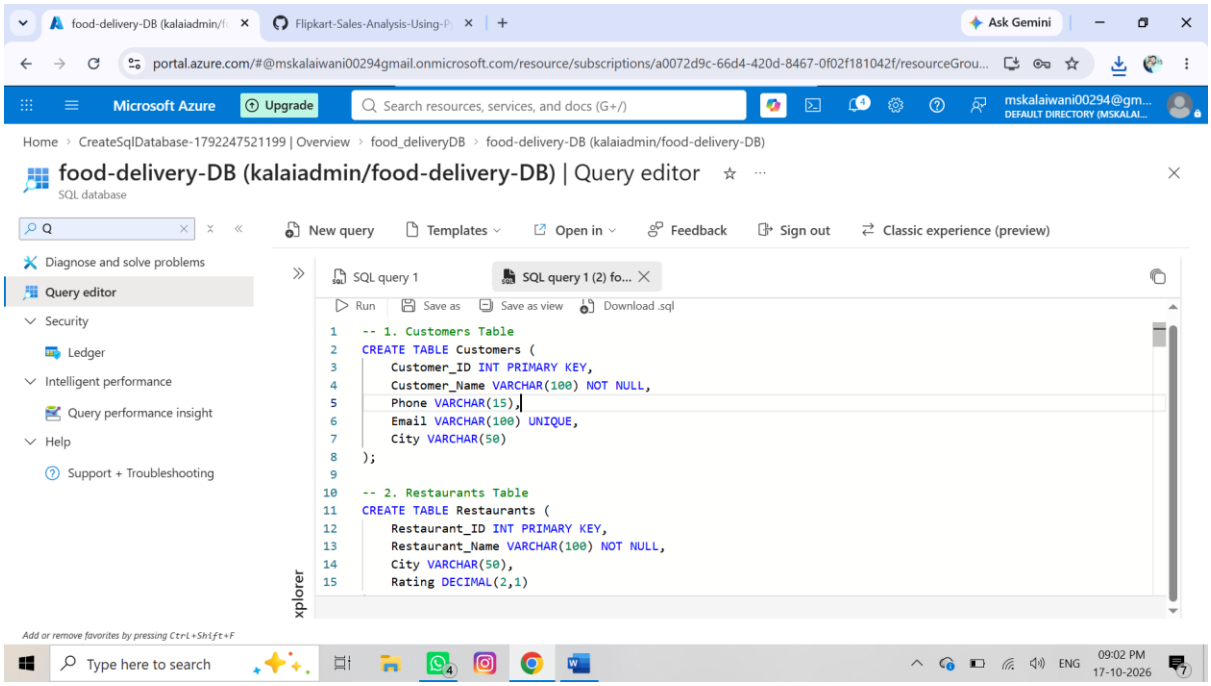


Which order has the highest total amount?



Which restaurants have the highest average rating?





My Analysis

I performed the following business analyses using SQL:

1. Total Orders: Counted the total number of orders to understand order volume.

2. Total Revenue: Calculated revenue from delivered orders to evaluate business performance.

3. Popular Food Items: Identified frequently ordered food items using JOIN and GROUP BY.

4. Customer Analysis: Analyzed customer order counts to identify frequent customers.

5. Restaurant Analysis: Compared restaurants based on the number of food items they offer.

6. Order Status Analysis: Examined order statuses to understand delivered, preparing, and cancelled orders.

7. Highest-Value Order: Identified the order with the highest total amount.

8. Restaurant Ratings: Compared restaurant ratings to understand customer preferences.

These queries help understand ordering patterns, customer behavior, restaurant offerings, and revenue performance.

Conclusion:

This project helped me understand how SQL can be used to analyze food delivery data and answer business questions. I practiced creating relational tables, defining primary and foreign keys, inserting data, joining tables, and using aggregate functions.

The analysis can help a food delivery business identify popular food items, understand customer ordering patterns, monitor order statuses, and evaluate revenue. This project improved my practical SQL skills and my ability to derive business insights from data.